

Workload Classification in Multi-VM Cloud Environment Using Deep Neural Network Model

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Abstract

In this competitive world, everyone needs to be prepared for future risks and emergency conditions. In a multi-cloud environment users can easily shift from one cloud to another cloud because of the available data and application transfer technologies. Therefore a strong forecast system is mandatory for such conditions and to stop user migration to other clouds. Virtual Machine (VM) plays an important role in effective resource management and cost reduction in cloud infrastructure. Workload prediction in multi-VM is very useful to handle uncertain situations. In this paper, we propose a promising workload prediction technique that can handle the workload from multiple virtual machines. It has a pre-processing and feature selection engine that handles direct data from these virtual machines and the model is strong enough in classifying data based on historical workloads. This classification enables extra knowledge for the cloud vendor to optimize resource usage. This strategy can be used for producing an alarm whenever there is continuously high utilization of resources in the future. Here, our prediction methodology is experimented with a popular real-world Grid Workload Archive (GWA) dataset and it achieves more than 85% prediction accuracy for CPU, Memory and Disk Utilization.

Keywords: Virtual Machine, Neural Network, Grid Workload Archive, Long Short Term Memory

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1 Introduction

The strategy to use prediction has led to the prevention of many emergencies and has induced a better control management over various complex systems. In the multi-cloud environment, the customer always looks for high availability, low cost, better latency, and demanded computation power. To fulfill this requirement effectively, the vendor has to look after the management of resources and better planning in advance for future critical situation. Such circumstances can be better handled by using proper prediction methodology for workload prediction. These predictions can become complex when there are a large number of users resulting in a high number of virtual machines (VMs). Thus there is a need for a suitable model to overcome the uncertain situations which affect the performance and the user's experience.

A time series data is a variable whose value is a set of observations at different times. But there can be much complex time series called multi-variate time series where getting the right information out of it can be very tough. Hence there is a need for an accurate prediction model according to the problem. Workload prediction in the virtual machine is a time-series data based prediction whose features can be CPU, Memory, Network, etc. usage at different intervals of time. Other than forecasting the workload, time series data are used in various fields of analysis, varies from weather readings to psychological signals. A time series is a data point sequence in a time domain, usually at a fixed interval. Everyone is switching to the cloud these days for various kinds of services and thus increasing the number of customers, utilizing more resources, making it difficult to provision the resources. Thus, an efficient method for workload prediction method that can be used for handling the complex resource allocation is required.

The paper is organized as follows. Section II discusses the work that has already accomplished in this field of cloud workload prediction. Section III describes the model algorithm in detail and provides a description of the methodology used for the prediction. Section IV provides a complete experiment evaluation and analysis report. Finally, section V, we draw our conclusion and mention some future work.

2 Related Works

Resource Provisioning in cloud can be immensely improved by using workload prediction. There has been some significant research in this area but there is still some scope

for improvement. There are some important techniques for workload prediction, that have been proposed over time. They can be classified into two as elaborated below.

2.1 Prediction based on Statistics

The statistical models are based on mathematical functions. The classical models use functions of auto-correlation and time series exponential decay. The models under this are Auto Regression (AR) and Moving Average (MA). In the AR model the random variable are expressed with the linear combination of the variables present in the preceding moments and thus the variable shift is done by itself. This model uses the sliding window to extract the time series features in case of adjacent data segment. There are also hybrid methods available, some of them being ARMA and ARIMA, where ARIMA is typically used for non-stationary regressive model. Rodrigo et al. [3] have proposed a model (ARIMA) for predicting workload, specifically the impact on Quality of Service (QoS) cloud applications.

Use of these models has also found its application by Yang et al. [10] where, they have used this models for auto-scaling of cloud service based virtual resources at different hierarchical levels. Liu et al. [6] proposed a model based on adaptive prediction for discrimination in cloud workload patterns. Amiri et al. [1] proposed a cloud workload prediction approach by using an online learning model based on episode mining. The proposed model adapts dynamically to changes in the workflow without additional data on the new workflows being required and directly extracts the pattern of behavior.

The statistical models are based on processing and transformations of stationary data. These methods might give a decreasing accuracy with Long-term and as complexity increases whereas their performance is good for linear short term prediction. Thus, the search for prediction solutions for non-stationary items is increasing.

2.2 Prediction based on Machine learning

Machine Learning is one of the rapidly growing area for classification and regression. The neural Back-propagation Network (BP), the Neural Radio-Base Function network (RBF), the Non-linear Auto-regressive Neural network (NAR), the Support Vector Machine (SVM) and the Bayes network have been studied and applied in the prediction problems.

Shyam et al. [9] proposed a machine learning based Bayesian model for predicting the resource requirement, specifically, this is related to long term and short term virtual machine requirement for a CPU / memory intensive application. Zhang et al. [12] use a Deep Learning based Model on the canonical polyhydrate decomposition primarily for predicting cloud workload for industry informatics, specifically workload prediction of virtual machine. Another work by Zhang et al. [13] uses Deep Belief Network (DBN) for resource request prediction. Kumar et al. [5] propose an Artificial Neural Network

and Adaptive Differential Evolution based workload prediction methodology for cloud. There are some regression-based studies have been conducted on prediction such as by [11] who use a linear model to predict workload. The data can be multi-variate time series where [2] uses primary network and auxiliary network as a compound auto-regressive model, designed to predict the multivariate series.

There is another work by Rahmanian et al. [7] dealing with cloud computing resource utilization estimation and uses ensemble model. In one of the papers, Hirwa et. al [4] propose a multivariate ensemble model for resource performance prediction in the cloud based on time series. Other than the linear model, many researchers have combined multiple techniques to predict workload. There is a work by Zhu et al. [14] where the author propose a technique based on attention mechanism with an RNN method and based on which LSTM encoder-decoder network is applied for workload prediction. After studying these mechanisms, it is found that ML models are well suited for multivariate prediction, thus an appropriate model is used for solving the problem of workload prediction. Specifically, Long Short Term Memory networks (LSTM) model with approximation approaches applied over the data is considered for the prediction model.

3 The Proposed Prediction Model based on Deep Neural Network

Machine Learning has several approaches under its belt with a particular collection of methods called Deep Learning Models. These are specific approaches with a definite architecture based on Neural Networks (NN) and are in many cases powerful than other traditional ML approaches. There are further several variants of these Deep Learning Models and every variant has its advantages and disadvantages, primarily based on the application for which it is to be used. There are deep learning models that suit a particular problem and works effectively in that environment. Hence choosing an appropriate model is a challenging problem in machine learning. Here we have chosen Long Short Term Memory Model which suits our current problem to be solved.

The proposed system implements the concept of a multi-layered LSTM network in the GWA dataset [8] for application in workload prediction in a multi-VM cloud environment. The system is processed in two phases: In the first phase, it selects the appropriate data and approximation method, and in the next phase it goes for LSTM model to perform training on historical data. Overall the system learns from the most important features, from multiple VMs and predicts the next class of future workload. The system is described in detail below.

The time-series data may contain many fluctuations and there may be high variance in data. There we need some method to avoid uncertain peaks to get knowledge of the right feature out of long time series. In multivariate time

series, the problem can be more complex and without an approximation, a model can take more time in training. Therefore, proper approximation reduces training time.

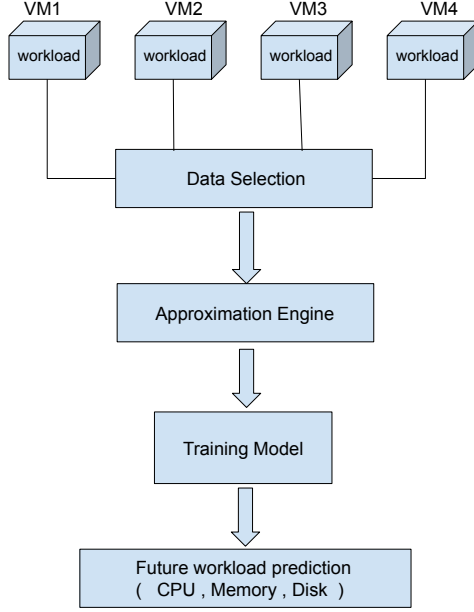


Figure 1. Prediction System

From Figure 1 there are multiple VM running on cloud infrastructure. Each VM has a workload parameter associated with CPU, Memory, Disk utilization history. The workload data then accumulated from each VM and pre-processed properly to be used by the next level. Further, the approximation engine again pre-processes it and finds the best feature out of whole data. The approximation engine computes the average value of a given segment (particular period) and replaces all the existing values of that segment with the average value. The labels selected from the data are discretized into 4 classes as High, Medium, Low, and Very Low.

Now the data from the approximation engine is fed into the model for training. Here the model uses a multi-layer LSTM network followed by dense layer, to learn the classification from features to labels. LSTM layers are very effective in obtaining the temporal relationship. This network tries to learn the local temporal relation between the past resource utilization within a similar timestamp and also the relationship with multiple virtual machines for the same timestamp. The network is trained to classify the workload usage for each virtual machine.

4 Experimental results and Analysis

Our goal is to predict future CPU, Memory, and Disk utilization for all virtual machines based on past resource utilization trends.

4.1 Data Pre-processing

We first perform a feature selection onto the raw data as this time series containing workload parameter samples for every 300ms because not all the features have the same importance to participate in prediction. There are some features which are not useful and are removed. This process involves grabbing the important data for resource utilization on all of the 500 VMs. Finally the original raw VM trace data of 500 CSV files with roughly 8,600 rows of data converted into CSV files with roughly 8,250 rows of data with important features. After this, some inappropriate rows are dropped. Further, the final data is prepared based on the experimented window size of resource utilization for CPU, Memory, and Disk.

To formulate the output of the dataset for the classification problem, here the resource utilization of each row is categorized as High, Medium, Low, and Very Low based on approximations and then performed one hot encoding to categorise output by the Deep Neural Network model. Because the data consists of several recorded entries of VMs at an idle state consuming no resources, the model has been created that would fairly predict all the workload parameters.

4.2 Results

The proposed model is implemented in Google Colab using the Keras library programming tool. The model is evaluated using the real-time GWA time series dataset [8]. It is one of the best time-series datasets representing metrics of various workload parameters. The dataset contains virtual machine resource usage measurements from the cloud datacenter. This dataset is pre-processed and approximated to suit the proposed deep learning model. For validating or testing the model, 20 percent of data is used. The model is trained on LSTM with a batch size of 32 for 200 epochs. During the training, care is taken, so that the model does not overfit and underfit by using proper regularisation along with the right capacity of the network.

The trained model is then tested over the test data for CPU, Memory, and Disk utilization. The results for CPU predictions are shown in Figure 2 which provides a complete report of accuracy along with precision, recall, and f1-score for all the classes of workload. The accuracy achieved in the case of CPU prediction is 91%. The results of Memory predictions are shown in Figure 3 and the accuracy achieved in this case is 88%. Figure 4 shows the results for Disk prediction where the accuracy is 85%. The precision, recall, and f1-score in all the three cases are above 80% which shows that the model is not biased for predicting any one class of workload.

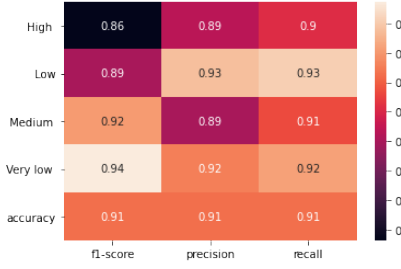


Figure 2. CPU utilization prediction

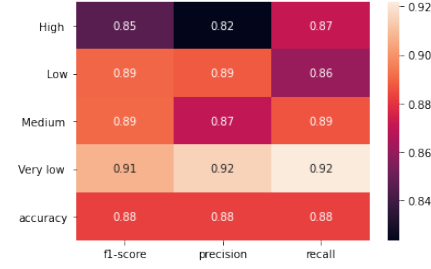


Figure 3. Memory utilization prediction

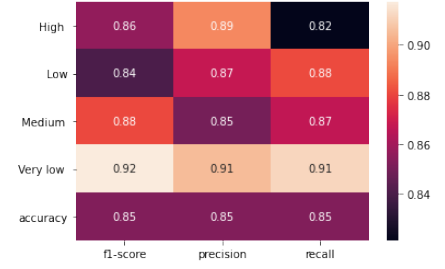


Figure 4. Disk utilization prediction

It is good enough to classify any class of workload data with high accuracy.

In this classification, due to the uneven distribution of each class, there is a need to consider precision, recall, and f1-score. From Figure 2, 3, 4 it can be confirmed from the results that this model's prediction for class 'Very Low' is better than other classes. And CPU prediction results for all four classes are better than Memory and Disk prediction. However, the overall results are good because the experiment is performed over 500 VMs running simultaneously [8]. All the results included here are an average of each resource prediction for all VM's. Overall the evaluation measures considered in the experiment show excellent results for the VM workload prediction.

5 Conclusion and Future Work

We employ the LSTM Model for Cloud based Virtual Machine Workload Prediction using a grid cluster dataset (GWA Dataset). The dataset itself is enormous, and additionally a time series dataset needs a lot of pre-processing and approximation for proper prediction. The pre-processing and data transformation effects the overall accuracy of the model. The purpose of this research is to classify the future workload occurrences which can be very useful for handling uncertain situations and here, the model is applied with appropriate pre-processing and approximation over the dataset. The models have been evaluated based on prediction results. All three workload parameters namely, CPU Utilization, Memory Utilization, and Disk Utilization, are considered. The calculated results indicate that the LSTM Model can be used for classifying workload in a Multi-VM environment. Training a model on LSTM over the data from each virtual machine takes more time and thus as a future work, we can try to decrease the training time by training the dataset in a distributed manner.

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